Completing the archive's `bitop/v1` accounting

`research/ic_acct6_archive_20261021` · study 2026-10-21 · tranche 1 run 2026-10-26 · report 2026-10-26 · class: accounting

Question

Of the archive's verified, non-warm-up ecbench records (every `research/**/records.jsonl`), how many price `complete` in `bitop/v1`, and how can the rest be made complete? `PROTOCOL.md` (frozen before any re-run) keeps two routes apart: **phase A**, new pricing rules for counters that committed records already carry, with no primitive weight changed; and **phase B**, re-running sessions on the instrumented binary to count work that was never counted. This is accounting. It changes no measured cost and is not an improvement of the index-calculus baseline.

Every number below is read from files written by the Rust example `examples/ic_acct6_archive.rs`: `breakdown-before.json`, `breakdown-after.json`, `plan.json`, `breakdown-tranche.json`, and the generated `BEFORE_AFTER.md`, `PLAN.md` and `TRANCHE.md`. The figures only plot them (`figures/plots.py`).

Phase A: new rules, no runs

Rules: `docs/units/BITOPS.md`, "Extensions", 2026-10-21 (rules at `42f23b5e4` against `4651c5d9f`); `committed_vectors_are_frozen` passes. Archive at main `cde2881dd`: 705 session directories, 119 unreadable both before and after, 73,012 verified non-warm-up records.

	before	after	change
records	73,012	73,012	0
complete	50,273	50,306	+33
incomplete (total is a lower bound)	22,682	22,706	+24
unpriceable (error)	57	0	−57

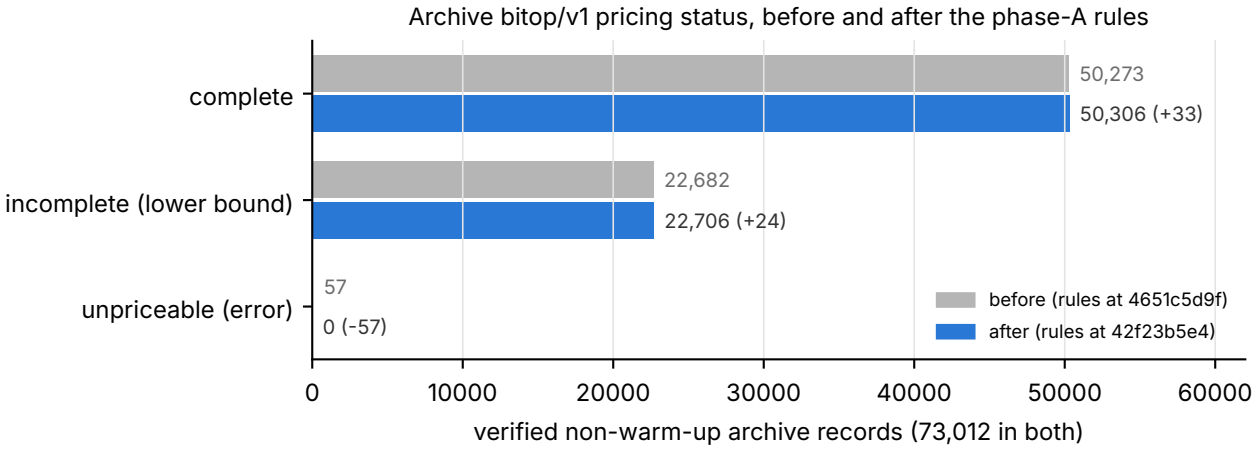


Figure 1: Pricing status of the 73,012 archive records before and after the phase-A rules (`breakdown-{before,after}.json`, `totals`). Exact counts.



Figure 2: The records the new rules touched, coloured by status after phase A. Extension-family rows from `by_method_family`; the CM-folded (`native_queries_uncharged`), projected-base (`projected_base_membership_checks_uncharged`) and isogeny-tower (`aux_supersingular`) rows from `by_reason` before (all three are 0 after).

All 57 extension-family records now price: 33 (`rho.negation`) complete and 24 (`ic.pipeline`) incomplete. The 628 projected-base, 96 isogeny-tower and 477 CM-folded records no longer name an unpriced counter but stay incomplete: the never-counted index-calculus items (`gauss_column_scans_and_pivot_inversions`, `relation_row_construction`, `table_insert_attempts`, `walk_bookkeeping`, `base_index_builds`) remain, and for CM-folded the native backend's unmetered work is now named as `uncounted`. The rules made these records honest about what is missing; they did not make them complete.

Phase B: the re-run plan

`PLAN.md`: 396 sessions hold an incomplete record (22,706 records, 71.55 hours of summed process wall time on the hosts that ran them). 337 are re-runnable unchanged (22,013 records, 48.58 h), 17 of them leaderboard sources (1,383 records, 12.73 h). 59 are not: CM-folded sessions whose pinned native backends are absent here, interrupted sessions with fewer records than their plan, and two without a frozen spec.

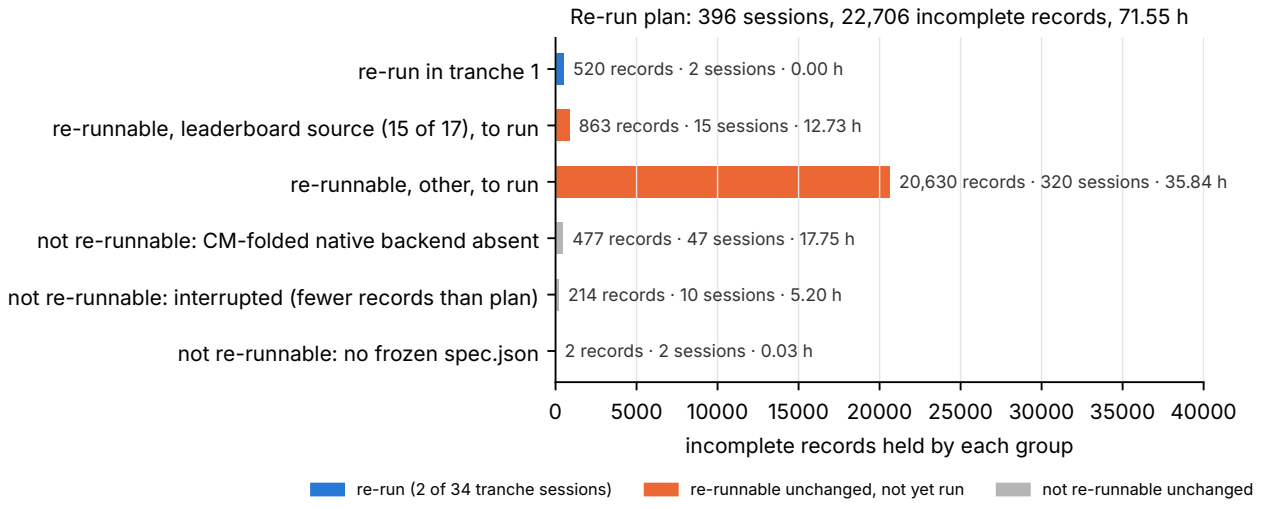


Figure 3: Incomplete records per re-run status group, from `plan.json` (396 rows; the groups count rows by their recorded `rerun` field and sum their recorded `incomplete` and `est_hours`; the script asserts PLAN.md's totals 396 / 22,706 / 71.55 h and 337 / 22,013). Hours are the original sessions' summed `process_wall_ns`, descriptive only.

Tranche 1: 2 of 34 sessions run

Card 1 never acquired the host's shared build lock and was stopped with no session started. Card 2 ran 07:16–10:20 UTC; the lock was free long enough for two sessions (22.8 s and 18.5 s of process wall time). The other 32 sessions of the tranche were not launched.

Both re-runs (`p001`, `p002`, the n37 `ic.shared_rank` leaderboard sessions) plan to the committed spec ids, and both audits pass (`ecbench verify --replay`
2: 384 records each, 2 replays reproduced, 0 problems). **Neither prices complete.** The current binary's `ic.shared_rank` writes only the `counted:base_index_builds` marker, so 520 of the 640 counted records still miss the four other never-counted items; the 120 complete records are the rho arms. The plan's `seen` column (0/6, 0/2) predicted this. No archive record is superseded.

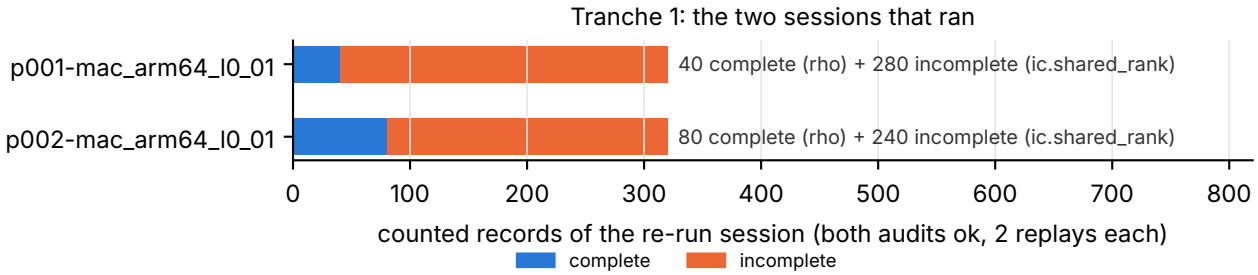


Figure 4: Counted records of the two re-run sessions by pricing status (`breakdown-tranche.json`, `sessions`). L0 runs (`--cpus 1 --allow-busy`) on a shared host; counts only.

Limitations and next step

- **The tranche barely ran:** 2 of 34 planned sessions, and both stay incomplete because `ic.shared_rank` is under-instrumented. Phase B has not yet completed a single archive IC record.
- **Recommended amendment** (not applied to the frozen protocol): run first the 24 re-runnable sessions whose incomplete configurations all have a complete record elsewhere (`seen` = total; 1,833 records, about 2.0 h), and instrument `ic.shared_rank`'s rank, row, walk and table paths before re-running its sessions, which include four of the leaderboard's first five.
- **119 unreadable session directories** (schema drift; listed in `BEFORE_AFTER.md`) are outside both breakdowns.
- **Validation gate.** No `VALIDATION.json`, no leaderboard or `benchmarks/` row: this is accounting, not a performance result.

Evidence

`PROTOCOL.md`, `BEFORE_AFTER.md`, `PLAN.md`, `TRANCHE.md`, `breakdown-{before,after,tranche}.json`, `plan.json`, `specs/`, `sessions/p00{1,2}-mac_arm64_l0_01/` (with `verify.json`), `logs/`, `cards/`, `run_tranche.sh`, `figures/plots.py` (plots recorded values only).